Statistics & Probability

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Mean, median, and mode**

Find the mean of $4, 8, 6, 10, 7$.

Sum $= 35$. Mean $= \frac{35}{5} = 7$.

2 **[Foundational]** • **Target: Mean, median, and mode**

Find the median of $3, 9, 5, 1, 7$.

Sorted: $1, 3, 5, 7, 9$. Median $= 5$.

3 **[Foundational]** • **Target: Mean, median, and mode**

Find the mode of $2, 4, 4, 6, 8, 4, 2$.

4 appears three times, more than any other value, so the mode is 4.

4 **[Foundational]** • **Target: Mean, median, and mode**

A data set has values $8, 12, 10, 15, 10, 14, 11$. Find the mean, median, and mode.

Sum $= 80$, mean $= \frac{80}{7} \approx 11.43$. Sorted: $8, 10, 10, 11, 12, 14, 15$; median (4th value) $= 11$. Mode $= 10$ (appears twice).

5 **[Foundational]** • **Target: Range and quartiles**

Find the range of $5, 12, 8, 20, 3$.

Range $= 20 - 3 = 17$.

6 **[Foundational]** • **Target: Range and quartiles**

For the data set $2, 4, 5, 7, 8, 9, 12$, find Q_1 and Q_3.

Median (4th value) $= 7$. Lower half $2, 4, 5$: $Q_1 = 4$. Upper half $8, 9, 12$: $Q_3 = 9$.

7 **[Foundational]** • **Target: Range and quartiles**

Find the interquartile range (IQR) for the same data set.

IQR $= Q_3 - Q_1 = 9 - 4 = 5$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Variance and standard deviation

Find the variance of $2, 4, 6, 8, 10$, using $\sigma^2 = \frac{\sum(x-\mu)^2}{n}$.

Mean $= 6$. Squared deviations: $16, 4, 0, 4, 16$, sum $= 40$. Variance $= \frac{40}{5} = 8$.

9 [Intermediate] • Target: Variance and standard deviation

Find the standard deviation of the same data set, rounded to two decimal places.

$\sigma = \sqrt{8} \approx 2.83$.

10 [Intermediate] • Target: Variance and standard deviation

A data set has variance 25. State its standard deviation.

$\sigma = \sqrt{25} = 5$.

11 [Intermediate] • Target: Frequency tables

A frequency table shows value 1 with frequency 3, value 2 with frequency 5, and value 3 with frequency 2. Find the mean.

Sum $= 1(3) + 2(5) + 3(2) = 3 + 10 + 6 = 19$. Total frequency $= 10$. Mean $= \frac{19}{10} = 1.9$.

12 [Intermediate] • Target: Frequency tables

Using the same frequency table, find the mode.

Value 2 has the highest frequency (5), so the mode is 2.

13 [Intermediate] • Target: Frequency tables

Using the same frequency table, find the median.

There are 10 values in order $1, 1, 1, 2, 2, 2, 2, 2, 3, 3$. Median $= \frac{2+2}{2} = 2$ (average of the 5th and 6th values).

14 [Intermediate] • Target: Basic probability

A fair six-sided die is rolled. Find $P(\text{number} > 4)$.

Favorable outcomes $\{5, 6\}$: $P = \frac{2}{6} = \frac{1}{3}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Basic probability • Cross-Lesson: 8.01 Exponentials & Logarithms

A bag contains 5 red and 3 blue marbles. Find $P(\text{blue})$.

$P(\text{blue}) = \frac{3}{8}$.

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16 [Challenging] • Target: Basic probability • Cross-Lesson: 1.01 Algebra & Equations

A card is drawn from a standard 52-card deck. Find $P(\text{king})$.

$P(\text{king}) = \frac{4}{52} = \frac{1}{13}$.

17 [Challenging] • Target: Basic probability • Cross-Lesson: 2.01 Functions & Transformations

Two fair coins are flipped. Find $P(\text{both heads})$.

$P = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$.

18 [Challenging] • Target: Complementary and mutually exclusive events • Cross-Lesson: 3.01 Sequences & Series

A die is rolled. Events $A = \{\text{even}\}$ and $B = \{\text{rolling a } 5\}$. Are A and B mutually exclusive?

Yes — $\{2, 4, 6\}$ and $\{5\}$ share no outcomes.

19 [Challenging] • Target: Complementary and mutually exclusive events • Cross-Lesson: 4.01 Trigonometry

If $P(A) = 0.4$ and $P(B) = 0.25$, and A, B are mutually exclusive, find $P(A \text{ or } B)$.

$P(A \text{ or } B) = 0.4 + 0.25 = 0.65$.

20 [Challenging] • Target: Independent events • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

Two independent events have $P(A) = 0.6$ and $P(B) = 0.5$. Find $P(A \text{ and } B)$.

$P(A \text{ and } B) = 0.6 \times 0.5 = 0.3$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: MCQ — Frequency tables

A frequency table shows value 1 with frequency 4, value 2 with frequency 6, and value 3 with frequency 5. Find the total number of data points.

a. 10

b. 15 ✓

c. 3

d. 6

$4 + 6 + 5 = 15$.

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22 [Foundational] • Target: MCQ — Frequency tables

Using the same frequency table, find the mean.

a. 2.33

b. ≈ 2.07 ✓

c. 2.00

d. 1.87

Sum $= 1(4) + 2(6) + 3(5) = 31$; mean $= 31 \div 15 \approx 2.07$.

23 [Foundational] • Target: MCQ — Frequency tables

The bar chart shows a frequency table. Find the total number of data points.

a. 15 ✓

b. 6

c. 3

d. 10

$4 + 6 + 5 = 15$.

24 [Foundational] • Target: MCQ — Conditional probability

A box has 12 balls: 7 red and 5 blue. Two balls are drawn without replacement. Given the first is blue, find P(second is blue).

a. $\frac{5}{11}$

b. $\frac{4}{11}$ ✓

c. $\frac{5}{12}$

d. $\frac{4}{12}$

After removing one blue, 4 blue remain out of 11 total.

25 [Foundational] • Target: MCQ — Counting: permutations and combinations

In how many ways can 6 people be arranged in a line?

a. 720 ✓

b. 120

c. 36

d. 6

$6! = 720$.

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26 [Foundational] • Target: MCQ — The five-number summary

For the data set $10, 15, 20, 25, 30, 35, 40$, find the median.

a. 25 ✓

b. 30

c. 20

d. 27.5

With 7 values sorted, the median is the 4th: 25.

27 [Foundational] • Target: MCQ — Reading data from a bar chart

The bar chart shows the number of books sold each day at a bookstore over five days. Find the total books sold.

a. 87 ✓

b. 22

c. 77

d. 97

$12 + 18 + 15 + 22 + 20 = 87$.

28 [Foundational] • Target: MCQ — Reading data from a bar chart

Using the same bar chart, find the mean number of books sold per day.

a. 15

b. 18

c. 20

d. 17.4 ✓

$87 \div 5 = 17.4$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: MCQ — Reading data from a bar chart

Using the same bar chart, find the range of books sold.

a. 12

b. 22

c. 8

d. 10 ✓

$22 - 12 = 10$.

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30 [Intermediate] • Target: MCQ — Reading data from a bar chart

Using the same bar chart, on which day were the most books sold?

a. Wednesday

b. **Thursday** ✓

c. Tuesday

d. Friday

Thursday shows the tallest bar, 22 books.

31 [Intermediate] • Target: MCQ — Word problems: statistics and probability in context

A survey of 250 students found that 150 like math and 100 like science, with 40 liking both. Find the probability that a randomly selected student likes math or science.

a. 0.84 ✓

b. 0.6

c. 0.16

d. 1.0

$\frac{150+100-40}{250} = \frac{210}{250} = 0.84$.

32 [Intermediate] • Target: MCQ — Word problems: statistics and probability in context

A quality-control inspector finds that 3% of items from a production line are defective. If 500 items are inspected, how many are expected to be defective?

a. 150

b. 15 ✓

c. 3

d. 30

$0.03 \times 500 = 15$.

33 [Intermediate] • Target: MCQ — Word problems: statistics and probability in context

In a survey of 180 people, 108 prefer coffee over tea. Find the probability, as a percent, that a randomly selected person prefers coffee.

a. 60% ✓

b. 50%

c. 108%

d. 40%

$\frac{108}{180} = 60\%$.

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34 **[Intermediate]** • **Target: MCQ — Range and quartiles**

A code uses 3 distinct digits from 0 to 9, where order matters. How many possible codes are there?

a. 710

b. 730

c. 1000

d. 720 ✓

$10 \times 9 \times 8 = 720$.

35 **[Intermediate]** • **Target: MCQ — Variance and standard deviation**

[illegible] (G4). In a study of the number of hours high school students spend studying per week, the following data was collected and represented in a bar chart. If the mean number of study hours for all students is 20 hours, find the value of *x* which represents the study hours for Group 4 (G4).

a. 20

b. 22

c. **24** ✓

d. 26

[illegible] 4 [illegible]. [illegible] 20 [illegible]. [illegible] = ([illegible]) / ([illegible]) 20 = (14 + 22 + 18 + x) / 4 [illegible] 4: 20 * 4 = 14 + 22 + 18 + x 80 = 54 + x [illegible] 54 [illegible] x: x = 80 - 54 x = 26 [illegible] (G4) [illegible] 26 [illegible]. To calculate the mean, we sum the number of hours for all groups and divide by the total number of groups. In thi.

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36 **[Intermediate]** • **Target: MCQ — Frequency tables**

[redacted] The bar chart above shows the distribution of student scores on a test. Which score range did the largest number of students achieve?

a. 60-69

b. 70-79

c. **80-89** ✓

d. 90-99

[redacted] 80-89 [redacted] 26 [redacted] (14 [redacted] 60-69[redacted] 22 [redacted] 70-79[redacted] [redacted] 18 [redacted] 90-99). To find the score range achieved by the largest number of students, we examine the bar chart and identify the tallest bar. The bar representing the 80-89 score range is the tallest, indicating that 26.

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37 **[Intermediate]** • **Target: MCQ — Basic probability**

[illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] 100 [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]

38 [Challenging] • Target: MCQ — Complementary and mutually exclusive events • Cross-Lesson: 3.01 Sequences & Series

[illegible] 15 [illegible]. [illegible] 97.5% [illegible] k [illegible] k[illegible] ([illegible] 2 σ [illegible]). The scores on an Arabic vocabulary test are normally distributed with a mean standard deviation of 15 points. If 97.5% of students scored less than k points, what is the value of k? (Assume the normal distribution follows the empirical rule or use the 2 σ approximation for standard deviations).

a. $\mu + 1.5\sigma$

b. $\mu + 2\sigma$ ✓

c. $\mu + 2.5\sigma$

d. $\mu + 3\sigma$

[illegible] 97.5% [illegible] 68% [illegible] $\pm 1\sigma$ [illegible] 95% [illegible] $\pm 2\sigma$ [illegible] 99.7% [illegible] $\pm 3\sigma$ [illegible]. [illegible] 97.5% [illegible] $\mu + 2\sigma$. In a normal distribution, the percentage 97.5% indicates that the value lies at two standard deviations above the mean. 68% of the data lies within $\pm 1\sigma$ of the.

39 [Challenging] • Target: MCQ — Independent events • Cross-Lesson: 4.01 Trigonometry

[illegible] A [illegible] C[illegible] The bar chart below shows the number of students who won gold medals in different competitions. What is the total number of gold medals won by students in competitions A and C?

a. 30

b. **34** ✓

c. 40

d. 44

[illegible] A [illegible] C[illegible] [illegible] A [illegible] 14. [illegible] C [illegible] 18. [illegible] A [illegible] C [illegible] 14 + 18 = 32. To find the total number of gold medals in competitions A and C, we need to read the corresponding values for each competition from the bar chart. The number of gold medals in competition A is 14. The number of gold medals in competition C i.

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40 [Challenging] • Target: MCQ — Conditional probability • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

[illegible]. [illegible] The following bar chart shows the number of students who prefer different fruits. Which fruit is the most preferred?

a. [illegible]

b. [illegible] ✓

c. [illegible]

d. [illegible]

[illegible]. [illegible] The bar chart displays the number of students who prefer each fruit. We look for the fruit with the tallest bar, indicating the highest number of preferences. The bar for bananas is the tallest, with 25 students preferring bananas. Therefore, bananas are the most preferred fruit.

41 [Challenging] • Target: MCQ — Counting: permutations and combinations • Cross-Lesson: 6.01 Limits & Derivatives

[illegible] $\mu + \sigma$ [illegible] $\mu + 2\sigma$ Percentage of data between 1 and 2 standard deviations above mean?

a. **13.5%** ✓

b. 34%

c. 2%

d. 68%

[illegible] μ [illegible] $\mu + 2\sigma$ [illegible] $\frac{95\%}{2} = 47.5\%$. [illegible] 34%: $47.5\% - 34\% = 13.5\%$.

42 [Challenging] • Target: MCQ — The five-number summary • Cross-Lesson: 7.01 Integration & Applications

[illegible]: 2, 4, 6, 8, 10 [illegible] Sample standard deviation of {2, 4, 6, 8, 10}?

a. $\sqrt{10}$ ✓

b. $\sqrt{8}$

c. 3

d. $\sqrt{5}$

[illegible] $\bar{x} = 6$. [illegible]: $(2-6)^2 + (4-6)^2 + (6-6)^2 + (8-6)^2 + (10-6)^2 = 16 + 4 + 0 + 4 + 16 = 40$.
[illegible]: $s^2 = \frac{40}{5-1} = \frac{40}{4} = 10 \Rightarrow s = \sqrt{10}$.

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43 [Challenging] • Target: MCQ — Reading data from a bar chart • Cross-Lesson: 8.01 Exponentials & Logarithms

■■■ ■■■ $P(A) = 0.4$ ■ $P(B) = 0.5$ ■■■■■ ■■■■■■■■■■ ■■■■■■■■■■ ■■■ ■■■■ $P(A \cup B)$■

Find P(A or B) for independent events.

a. **0.7** ✓

b. 0.9

c. 0.2

d. 0.6

■■■ ■■■■■ ■■■■■■■■■: $P(A \cap B) = P(A) \cdot P(B) = 0.4 \times 0.5 = 0.2$. ■■■:

$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.4 + 0.5 - 0.2 = 0.7$.

44 [Challenging] • Target: MCQ — Word problems: statistics and probability in context • Cross-Lesson: 1.01 Algebra & Equations

■■ ■■■■ ■■■■■ ■■■■■■■■: 2, 5, 8, 10, 14, 18■ Find the median of an even number of values.

a. **9** ✓

b. 8

c. 10

d. 8.5

■■■■■ ■■■■■ ■■■■■■■ ■■■■■ (6 ■■■■)■ ■■■■■■■ ■■ ■■■■■■ ■■■■■■■■■■ ■■■■■■■■■■ (8 ■ 10): $\frac{8+10}{2} = 9$.

45 [Challenging] • Target: Mean, median, and mode • Cross-Lesson: 2.01 Functions & Transformations

■■■ ■■■■■ ■■■■■■■■■■ A ■ B ■■■■■■■■■■■ ■■■ $P(A \cap B)$ ■■■■■: If A and B are mutually exclusive, P(A and B) = ?

a. **0** ✓

b. 1

c. $P(A) \cdot P(B)$

d. $P(A) + P(B)$

■■■■■■■■■■ ■■■■■■■■■■■■ ■■ ■■■■■ ■■■■■■■■ ■■■■■ ■■ ■■■■ ■■■■■■■ ■■■■ ■■■■■■■ ■■■■■■■■■ ■■■■■■ 0.

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